1 error evaluation for propagation velocity

We use the four valid experimental phase velocities (the fifth point with $V_p = 0$ was discarded):

$V_{p,1} = 380.2281369\,\text{m/s}, V_{p,2} = 375.2345216\,\text{m/s}, V_{p,3} = 373.9249657\,\text{m/s}, V_{p,4} = 373.2736096\,\text{m/s}.$

1.1 Instrumental (propagated) uncertainty for each measurement

For a single measurement

$$V_p = \frac{\Delta x}{\Delta t},$$

the propagated uncertainty (first–order) is

$$\sigma_{V_p} = V_p \sqrt{\left(\frac{\sigma_{\Delta x}}{\Delta x}\right)^2 + \left(\frac{\sigma_{\Delta t}}{\Delta t}\right)^2}.$$

We adopt the typical instrumental precisions used previously:

$$\sigma_{\Delta x} = 1\ mm = 1 \times 10^{-3}\ \text{m}, \qquad \sigma_{\Delta t} = 0.1\ \mu s = 1 \times 10^{-7}\ \text{s},$$

and the nominal probe step $\Delta x = 1\ cm = 1 \times 10^{-2}$ m. Using the measured Δt values (in μs) from the table: $\Delta t_1 = 26.3\ \mu$s, $\Delta t_2 = 27.0\ \mu$s, $\Delta t_3 = 26.93\ \mu$s, $\Delta t_4 = 26.93\ \mu$s, the relative contributions are

$$\frac{\sigma_{\Delta x}}{\Delta x} = 0.100, \qquad \frac{\sigma_{\Delta t}}{\Delta t} \approx \{\, 0\ .00380(\Delta t_1)0.00370(\Delta t_2)0.00371(\Delta t_3,\, \Delta t_4)$$

so the combined factor is $\sqrt{0.100^2 + (0.0037{-}{-}0.0038)^2} \approx 0.10007$. Thus the per-measurement propagated uncertainties are approximately

$\sigma_{V_p,1} \approx 0.10007 \times 380.2281 \approx 38.05\,\text{m/s}, \sigma_{V_p,2} \approx 0.10007 \times 375.2345 \approx 37.56\,\text{m/s}, \sigma_{V_p,3} \approx 0.10007 \times 373.9250 \approx 37$

(These values show that the instrumental uncertainty for a single measurement is $\sim 10\%$ of V_p.)

1.2 Average and statistical uncertainty

The sample mean of the four phase velocities is

$$\bar{V_p} = \frac{1}{4}\sum_{i=1}^{4} V_{p,i} = 375.6653085\ \text{m/s}.$$

The sample standard deviation (unbiased) is

$$s \approx 3.15\ \text{m/s},$$

so the standard error of the mean (statistical uncertainty) is

$$\sigma_{\bar{V}_p}^{(\text{stat})} = \frac{s}{\sqrt{N}} = \frac{3.15}{2} = 1.575\ \text{m/s}.$$

1.3 Reduction of instrumental uncertainty by averaging

Assuming the instrumental (propagated) uncertainties for different measurements are independent (random measurement/readout errors), the instrumental uncertainty on the mean is reduced. A convenient estimator is

$$\sigma_{\bar{V}_p}^{(\mathrm{instr})} = \frac{\sqrt{\sum_{i=1}^{N} \sigma_{V_p,i}^2}}{N}.$$

Using the $\sigma_{V_p,i}$ above:

$$\sqrt{\sum_{i=1}^{4} \sigma_{V_p,i}^2} \approx 75.17 \text{ m/s}, \qquad \sigma_{\bar{V}_p}^{(\mathrm{instr})} \approx \frac{75.17}{4} = 18.79 \text{ m/s}.$$

(If one assumes identical instrumental uncertainty for each measurement, the simpler formula $\sigma_{\bar{V}_p}^{(\mathrm{instr})} \approx \sigma_{V_p}/\sqrt{N}$ gives a comparable value: $\approx 37.6/\sqrt{4} = 18.8$ m/s.)

1.4 Combined uncertainty

We combine the (reduced) instrumental uncertainty on the mean and the statistical standard error in quadrature:

$$\sigma_{\bar{V}_p}^{(\mathrm{tot})} = \sqrt{\left(\sigma_{\bar{V}_p}^{(\mathrm{stat})}\right)^2 + \left(\sigma_{\bar{V}_p}^{(\mathrm{instr})}\right)^2} \approx \sqrt{1.575^2 + 18.79^2} \approx 18.86 \text{ m/s}.$$

1.5 Final result

The final reported ion acoustic phase velocity, including both statistical and reduced instrumental uncertainties, is

$$\bar{V}_p = 375.7 \pm 18.9 \text{ m/s} .$$

Remarks.

- The dominant contribution to the total error is the instrumental (propagated) uncertainty per measurement; averaging reduces this contribution roughly by a factor $\sim \sqrt{N}$.
- If the manipulator positioning uncertainty can be reduced (e.g. $\sigma_{\Delta x} = 0.1$ mm) or the timing precision improved, the instrumental term will shrink and the total uncertainty will approach the statistical one (~ 1.6 m/s).
- The numerical values above use the assumptions $\sigma_{\Delta x} = 1\ mm$ and $\sigma_{\Delta t} = 0.1\ \mu$s.